twelve hydrogen atoms, and the oxygen atoms of sixteen hydrogen atoms. The hydrogen atom would thus be the only, ultimate building block of all matter. The hypothesis which postulated the existence of nearly a hundred different elements, had always been regarded as rather difficult to accept. For if we really believe in homogeneity in nature, we must obviously prefer the number of basic elements to be considerably smaller.

Despite the captivating features of Prout's hypothesis, it remained almost completely neglected for more than a hundred years. The main reason for its being discarded was that the atomic weights of the heavier atoms had not been proved to be approximately whole numbers. Nevertheless, this hypothesis did actually contain a grain of truth of considerable importance. We shall see later on that, in a modified form, it plays a fundamental part in modern nuclear physics.

A new era in atomic theory was introduced by Faraday (1791–1867), who combined atomic theory with the theory of electricity. Atomic theory is indebted to him for the formulation of a law of prime importance: In chemical changes induced electrically—by electrolysis—there is always a relationship between the matter transformed and a definite quantity of electricity. Moreover, Faraday discovered that the masses of the substances transformed by a certain definite quantity of electricity, are related to the so-called 'equivalent weights,' and thus in the simplest cases—those of univalent substances—to the atomic weights of the substances concerned. This discovery indicated that electricity, too, possessed an atomic structure, of such nature that each atom, or each molecule, of a chemical compound was always associated with one or more atoms of electricity, even though in some manner then still unknown. This was pointed out by Weber as early as 1848. It explains logically why the same quantity of electricity was always associated with the same quantity—in other words, the same number of atoms—of a substance. To-day, the *mol* (also written *mole*) and the *gramme-atom* are customarily employed as basic mass units. One mol of any substance is that quantity which weighs as many grammes as the numerical value of its molecular weight; one gramme-atom of an element is a quantity equal in grammes to the numerical value of its atomic weight. Thus one mol of oxygen gas, O_2, (molecular weight, 32) is equal to 32 grammes of oxygen gas, and one gramme-atom of oxygen, 0, (atomic weight, 16) is equal to 16 grammesof oxygen.[b] Every gramme-atom of a univalent element is associated with a certain quantity of electricity, 96,520 coulombs, and one gramme-atom of a multivalent element is associated with the appropriate multiple of this quantity.

The next advances were made in the domain of the theory of gases, raised to the status of an exact science as a result of the efforts of Maxwell,